Supplementary Materials: When Design Decisions Go to the Agent — A Governance Experiment in a Virtual Battery Factory

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Appendix A. Mechanical Re-Adjudication Rules (Control Conditions)

The re-adjudication of the control conditions applies the protocol's own definitions mechanically, by recomputing from artifact files. The rules:

- **Plating.** A design plates if the minimum of the anode surface potential time series (vs. Li/Li^+) during the 4C charge is below 0 V. A claim of "safe margin" phrased in kinetic overpotential is not a substitute for the potential series; if no potential series exists in the artifacts, the criterion is *unadjudicable* and counted as failed.

- **5C retention.** Computed as the 5C discharge capacity divided by the 1C discharge capacity of the same cell ($5C/1C$). A ratio computed against a $C/3$ reference is a different caliber and is recomputed.

- **SEI thickness.** Judged on the contract aging model (Chen2020, ec-reaction-limited SEI kinetics) with the design's full declared parameters. A design that reports SEI from a different growth law (e.g., the solvent-diffusion-limited variant) is recomputed under the contract model—the SEI model option is a contract-caliber fixture, not a design lever.

- **Thermal runaway / nail penetration.** Judged by the trigger criterion of the three-side-reaction ODE solver (`run-tr`): `triggered` must be `False`. A steady-state hotspot temperature is not a trigger criterion.
- **Measurement space.** A design is out-of-space if its chemistry or model is not reproducible by the protocol's verified parameter sets (Chen2020, ORegan2022, OKane2022, Prada2013); such designs are documented separately and excluded from the same-space attainment counts.

Appendix B. C1 Re-Adjudication Detail

Table B.1 restates the main-text table with the exact values and the failing criterion per task, and Figure B.1 shows the mechanical plating criterion applied to the T2 design; Figure B.2 quantifies the SEI-kinetics bridge on the same task.

Appendix C. Task–Product Anchoring Table

The thresholds in the task texts are calibrated from the underlying parameter sets at the DFN contract caliber (mechanically derived via `calc-energy`); the anchoring is a Methods-section record and never appears in the task text seen by any agent. Table C.2 lists the anchoring source per task.

Appendix D. Contract Margins of the Governed Runs

Margins are computed mechanically from the evidence chains of the final passing evaluations: margin $= (v - t)/|t|$ for minimum thresholds and $(t -$

Table B.1: Protocol-free agent claims vs. contract-caliber re-adjudication.

Task	Self-reported	Re-adjudicated value	Failing criterion
T1	"safe" plating margin	min $\eta = -6.9$ mV	< 0 V $\Rightarrow$ plated
T2	SEI 8.1/15.1 nm "all verified"	818.9 nm @500 cyc (contract model, full design); 4C plating -360.9 mV	SEI ≤ 550 nm @500 cyc; no plating
T3	5C retention 96.7%	84.8% ($5C/1C$)	retention $\geq 95\%$
T4	achieved (Li-metal cell)	—	out-of-space
T5	$\|\eta\|$ "3–4× safe"	min $\eta = -12.6$ mV	< 0 V; no series
T6	(no deliverable)	—	budget exhausted
T7	"all met"	hotspot 118.5 °C	no trigger criterion
T8	achieved (Li-metal cell)	—	out-of-space

$v)/|t|$ for maximum ones, with v the evidence value and t the pre-registered threshold. Boolean criteria (plating, thermal-runaway trigger) pass exactly and carry no margin. Table D.3 gives the complete per-criterion spreadsheet.

BO seed robustness. The Bayesian baseline is deterministic in its seed,

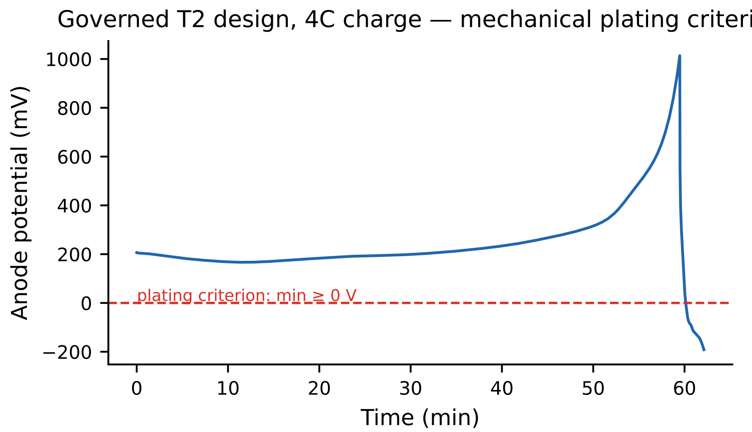

Figure B.1: The mechanical plating criterion as applied: anode potential time series of the governed T2 design during 4C charge, with the 0 V line. A design plates if and only if the minimum crosses this line.

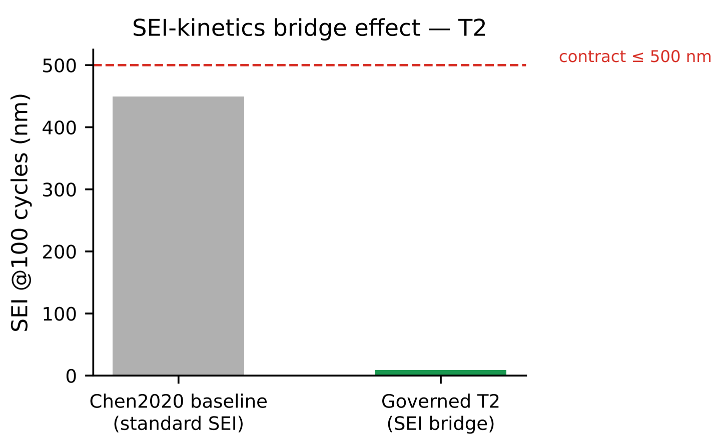

Figure B.2: SEI-kinetics bridge effect on T2: Chen2020 baseline SEI after 100 cycles versus the governed design's bridged value, against the 500 nm contract line.

and a three-seed replication (1234, 5678, 9012) leaves the attained-task set unchanged. Attaining-trial counts per task and seed (out of 50 evaluations) are tabulated in Table D.4:

Table C.2: Scenario anchoring and threshold calibration sources.

Task	Anchored scenario	Calibration source
T1	LG M50T fast-charge EV (4C + thermal red line)	ORegan2022 contract-caliber baseline 327.18 Wh/kg $\rightarrow$ target 392.61 (+20%)
T2	Grid storage (cycle + low temperature)	OKane2022 100-cycle SEI baseline 626.8 nm $\rightarrow$ 500 nm limit; Chen2020 low-T retention 99.5% $\rightarrow$ 90% limit
T3	Power-tool high-rate	OKane2022 5C/1C $\approx$ 100% $\rightarrow$ 95%; power baseline 3814 $\rightarrow$ 4000 W/kg
T4	Extreme-cold equipment	OKane2022 volumetric baseline 854 $\rightarrow$ 880 Wh/L
T5	Flagship extreme verification	Chen2020 contract baseline 400.75 $\rightarrow$ 500.94 Wh/kg
T6	Smartphone (volumetric + temperature red line)	OKane2022 854 $\rightarrow$ 950 Wh/L; NMC811 plateau limit 4.03 V $\rightarrow$ 4.1 V
T7	Hybrid (high-temperature + nail)	45 °C 100-cycle SEI $\leq$ 550 nm; 10 W nail penetration
T8	Long-endurance drone	OKane2022 baseline 405.62 $\rightarrow$ 446.18 Wh/kg; mass $\leq$ 40 g

The SEI criteria are adjudicated on the contract aging model (Chen2020, ec-reaction-limited), in which the electrode-modification bridge scales the

SEI kinetic rate constant k_{SEI} (standard 10^{-12} m/s). The governed runs used this lever as follows. T2 shipped $k_{\mathrm{SEI}} = 10^{-16}$ m/s ($10^{-4}\times$), declared in the audit as an estimate beyond the bare-ALD literature magnitude under a dual-mechanism coating claim (ALD plus fluorinated-additive inner SEI); the same design at 10^{-14} m/s ($10^{-2}\times$) meets the identical contract, with the 500-cycle SEI at 500.4 nm against the 550 nm limit (a 9% margin) so the T2 SEI criteria do not hinge on the extreme setting. T6's SEI work uses a $0.5\times$ solvent-diffusivity and concentration cut, derived from the model's own diffusion-limited rate law, and no T6 verdict depends on SEI. T7 uses no kinetic override; its 465.5 nm SEI follows from the design itself.

Appendix E. Real-Data Alignment: Source and License

The real-data anchor uses the LG M50 battery degradation dataset (Imperial College London), Zenodo record `10.5281/zenodo.10637534`, licensed under CC BY 4.0, accompanying the paper by Kirkaldy et al. (10.1016/jpowsour/2024.234185). The comparison is against Experiment 5 (standard cycle aging), Cell A, RPT0 (beginning of life), 0.1C discharge.

Appendix F. Replication Runs (N=3, All Eight Tasks)

Three seeds of every task under the governed protocol, deepseek-v4-pro, same budget as the main matrix (Table F.5).

All twenty-four runs achieve their contracts under the mechanical adjudicator, with complete evidence coverage of every pre-registered criterion in every run. On T6, all three seeds independently reach the same material-bottleneck diagnosis and solution (the LNMO system switch) through three different designs. Contract attainment and the bottleneck diagnosis are reproducible across seeds; the designs themselves differ.

Appendix G. Positioning vs. Battery-Sim-Agent

Battery-Sim-Agent [1] is the closest published system; Table G.6 positions the two papers side by side.

Appendix H. Reproducibility Statement

All simulation protocols, the adjudication layer, and the audit-chain verifier are distributed with the tool library and pinned by an automated regression suite. The full code base (protocol, tool library, experiment harnesses (including the cross-harness legs)) is open source at `https://github.com/wangyuanxty/BatteryDesignAgent`. The audit ledgers (`log.jsonl`) of every run are the primary data source of the paper; every table value can be re-derived mechanically from the ledgers and the tool outputs they reference. LLM runs are not seed-reproducible; the protocol's verdicts are.

Appendix I. The T6 Case: Candidate Sheet, Datasheet, and Curves

The T6 walk-through in the main paper (Section 5.6) compresses the run's materials to their decision structure. This section carries the detail. Table I.7 reproduces the run's full candidate sheet: twenty proposals, each logged with its name, type, and one-line rationale; the sequence reads like a design engineer's notebook: thermal probes first, then plating fixes on three distinct levers (anode capacity headroom, anode kinetics, electrolyte transport), then SEI levers, a combined candidate, DFN confirmation, and finally DFN-gap fixes with a margin-widening probe. Table I.8 condenses the technical datasheet of the final design (VBF-T6R1-DSH-01), whose Source column points each value at its tool-output file. Figure I.3 plots the decision trajectory with the contract line and the baseline ceiling; Figure I.4 shows the two decisive curves—the plateau gap that forced the system switch and the plating margin the criterion judges.

The datasheet closes with a formulation note that a production engineer would demand and an ungoverned agent would omit: the electrolyte transport numbers are design targets at the upper bound of reported fast-charge formulations, and "materializing the design requires a real low-viscosity electrolyte with measured D_e in this range": the deliverable labels its own estimates.

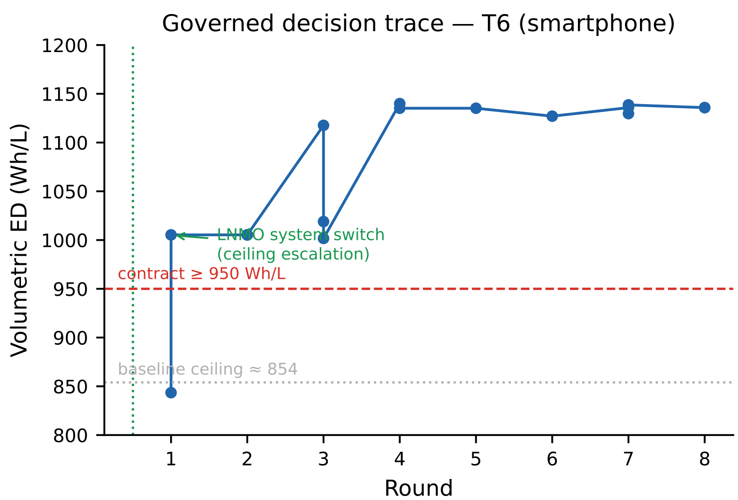

Figure I.3: Governed decision trace on T6 (smartphone contract): volumetric energy density by round, with the contract line (950 Wh/L) and the baseline ceiling (854 Wh/L). The jump across the dashed line is the ceiling escalation (the LNMO system switch) followed by rounds of refinement to 1135.8 Wh/L.

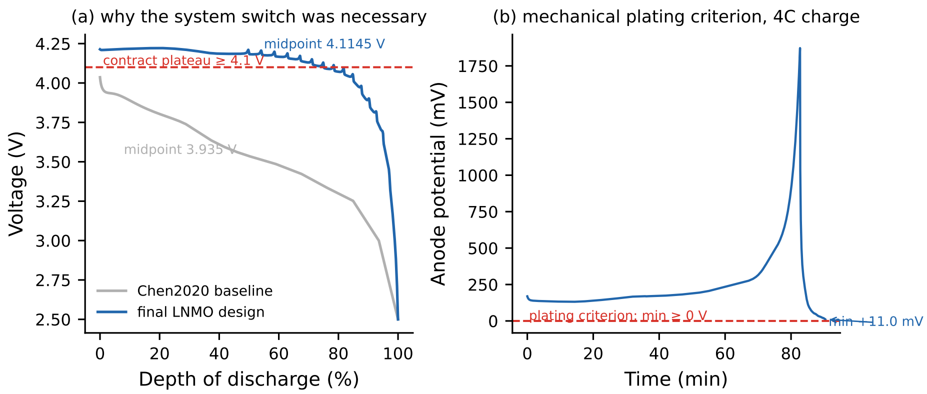

Figure I.4: The T6 case at curve level. (a) 1C discharge: the Chen2020 baseline's midpoint voltage sits below the 4.1 V contract plateau, which is what the ceiling assessment diagnosed; the final LNMO design clears it (4.1145 V). (b) The final design's anode potential during 4C charge stays above the 0 V plating criterion throughout (min +11.0 mV).

Appendix J. Threshold Sensitivity of the Baseline Attainment

The contract thresholds are author-anchored (fixed increments over the anchoring parameter sets' baselines), so the reader's natural question is how far the comparison depends on where exactly the thresholds sit. Table J.10 recomputes the Bayesian-optimizer attainment count under a uniform ±5% perturbation of every numeric criterion, directly from the stored 50-trial trajectories (no new runs).

Two structural properties follow. *First*, the five material-bottleneck tasks are threshold-invariant at ±5%: the optimizer's failure there is ceiling-bound, not threshold-bound. Across all 400 trials, the highest midpoint voltage any design reaches is 4.0115 V against the 4.1 V contract line; plating and SEI violations are lever-bound, so loosening the numeric thresholds cannot admit a design that is physically outside the parameter space's reachable set. *Second*, on the three structural tasks the same criteria bind both conditions in parallel: the governed agent's binding margins on T1/T4/T8 are +2.2%/+4.5%/+0.5% (Table D.3), so a uniform +5% tightening fails the governed runs on the very categories it fails the optimizer. The direction of the comparison therefore never inverts under a $\pm 5\%$ perturbation; the only possible inversion zone is energy-density thresholds set above the governed designs' own margins ($> +40\%$ on the structural tasks), which is exactly the region where the main text already reports the optimizer's structural-task energy-density superiority (Section 5.1).

References

[1] J. Chen, X. Gui, S. Fang, S. Tao, S. Zheng, W. Liu, J. Bian, Battery-sim-agent: Leveraging llm-agent for inverse battery parameter estimation, in: Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2026. `doi:10.1145/3770855.3818856`.

Table D.3: Per-criterion margins of the eight governed runs (pro model). The left column lists the criterion each task cleared last (binding margin); the right column lists the wide-margin criteria.

Task	Binding criterion (value vs threshold)	Margin	Wide margins
T1	$T_{\max}$ 325.65 vs 333.15 K	+2.2%	ED 553.98 vs 392.61 (+41.1%)
T2	low-T retention 99.57 vs 90%	+10.6%	ED 465.62 vs 327.18 (+42.3%); SEI 9.1 vs 500 nm (+98.2%)
T3	$T_{\max}$ 329.77 vs 333.15 K	+1.0%	power 32 630 vs 4000 W/kg (+716%); 5C retention 97.53 vs 95% (+2.7%)
T4	low-T retention 99.27 vs 95%	+4.5%	Wh/L 925.8 vs 880 (+5.2%); ED 471.55 vs 327.18 (+44.1%)
T5	$T_{\max}$ 328.13 vs 333.15 K	+1.5%	ED 641.83 vs 500.94 (+28.1%)
T6	plateau 4.1145 vs 4.1 V	+0.4%	Wh/L 1135.8 vs 950 (+19.6%); $T_{\max}$ 321.42 vs 323.15 K (+0.5%)
T7	SEI 465.5 vs 550 nm	+15.4%	ED 483.25 vs 327.18 (+47.7%)
T8	mass 39.8 vs 40 g	+0.5%	ED 634.32 vs 446.18 (+42.2%); 5C retention 98.77 vs 90% (+9.7%)

Table D.4: BO contract-attaining trials per task across three seeds (mechanical criteria as in the main text; the attained-task set is $\{T1, T4, T8\}$ in every seed).

Task	seed 1234	seed 5678	seed 9012
T1	25	28	21
T2	0	0	0
T3	0	0	0
T4	34	33	32
T5	0	0	0
T6	0	0	0
T7	0	0	0
T8	10	5	1

Table F.5: Threefold replication of all eight tasks.

Task	r1	r2	r3
T1 (structural)	✓ 553.98 Wh/kg	✓ 583.2 Wh/kg	✓
T2 (durability)	✓ 465.62 Wh/kg	✓	✓
T3 (rate)	✓	✓	✓
T4 (cold)	✓ 471.55 Wh/kg	✓	✓
T5 (flagship)	✓	✓	✓
T6 (material)	✓ 1135.8 Wh/L	✓	✓
T7 (hybrid)	✓	✓	✓
T8 (drone)	✓	✓	✓

Table G.6: Positioning against Battery-Sim-Agent (KDD 2026).

	Battery-Sim-Agent	This paper
Task	inverse parameter estimation (200 synthetic calibration cases)	forward cell design (8 scenario contracts)
Objective	curve-matching error	multi-criterion contract attainment
Baselines	BO (Ax, seed 1234), CMA-ES, defaults	BO (Ax, seed 1234, same config), protocol-free agent
Adjudication	agent-reported error	code-mechanical verdicts + evidence chain
Headline	67–95% error reduction vs. BO	8/8 vs. 3/8 (BO) vs. 0/8 in-space (C1)
Honesty calibration	low-headroom: BO can tie or win	same low-headroom pattern, plus a yardstick-failure axis BO cannot exhibit

Table I.7: The T6 candidate sheet (condensed from the report's full list of twenty; each proposal is logged with its rationale in the ledger).

Round	Candidate	Purpose (logged rationale)
R01	systemLNMO	system switch: 4.7 V-class LNMO spinel
R01	baselineChen2020	anchor-table default baseline
R02	lnmo_h200 / h500	thermal probes at 4C ($h = 200/500$)
R03	np120 / nr35 / elyte	plating fixes: N/P headroom; anode kinetics; electrolyte transport
R03	sei6e13	SEI-suppression probe (electrode modification)
R04	np125 / np120_nr35	plating closure: N/P 1.25 and a leaner alternative
R04	sei_c2270_d12 / d10	SEI levers: EC concentration and D_{EC} cuts
R05	lnmo_final	combined final candidate
R06	lnmo_final	DFN confirmation of the combined candidate
R07	r7_h400_d5 / nr30 / combo	DFN-gap fixes: thermal; plating; combined
R08	r8_final / r8_d6_margin	final candidate and margin-widening probe

Table I.8: Technical datasheet, condensed (VBF-T6R1-DSH-01); the Source column of the original points each value at its tool-output file.

Field	Value
Rated capacity (1C, DFN)	6.20 Ah
Plateau (discharge midpoint)	4.1145 V (window 2.5–4.7 V)
Volumetric ED	1135.8 Wh/L
Gravimetric ED	493.3 Wh/kg (electrolyte excluded)
4C fast charge, 45 °C	T_{max} 321.42 K (48.27 °C), no plating (min +10.96 mV)
DC resistance / power	6.38 mΩ / 13.52 kW/kg
Cycle life	100 × 1C: SEI 385.3 nm (≤ 500 nm)
Mass	51.50 g (electrolyte excluded; ≈57.8 g with estimate)

Table I.9: Final design parameters (excerpt of the design specification, VBF-T6R1-DS-01).

Item	Value
System	LNMO high-voltage spinel / graphite (4.7 V-class)
Positive electrode	75.6 µm, porosity 0.335, particle 5.22 µm
Negative electrode	102.2 µm, porosity 0.25, particle 3.5 µm
Separator	12 µm, porosity 0.47
Current collectors	Al 16 µm / Cu 12 µm
Electrolyte	EC/EMC + $LiPF_6$; EC 2270.5 mol/m^3, $D_{\mathrm{EC}} = 1 \times 10^{-18}$ m^2/s (SEI lever)
Cooling	$h = 400$ W/m^{2}K

Table J.10: BO attained trials under uniform ±5% threshold perturbation (seed 1234, 50 trials per task; × = each numeric criterion tightened/loosened by 5%, boolean criteria unchanged).

Task	Baseline	+5%	−5%	Why the count moves (or not)
T1	25/50	0/50	27/50	thermal line of the safety exam binds
T2	0/50	0/50	0/50	plating; SEI bridge is material
T3	0/50	0/50	0/50	plating; 5C retention is material
T4	34/50	22/50	38/50	low-temperature retention binds
T5	0/50	0/50	0/50	plating; chemistry ceiling
T6	0/50	0/50	0/50	plateau ceiling: max 4.0090 V $<$ 4.1 V
T7	0/50	0/50	0/50	SEI; plating
T8	10/50	1/50	11/50	mass ceiling 40 g binds